IAPT demand forecasting model

Week-3

Task-1 (DS-1)

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1. Dataset Description:

For the forecasting we need **IAPT referral volume time series** data. There is no direct data on NHS website related to referral time series. So, we have to go to the monthly statistics on IAPT data in which we have the count of referral received. From that the data from **2023-2026** (present) is extracted.

	Month	Referrals_Received
0	Jan-23	163719
1	Feb-23	148669
2	Mar-23	167407
3	Apr-23	129753
4	May-23	146659

Shape of data:

```
(38, 2)
```

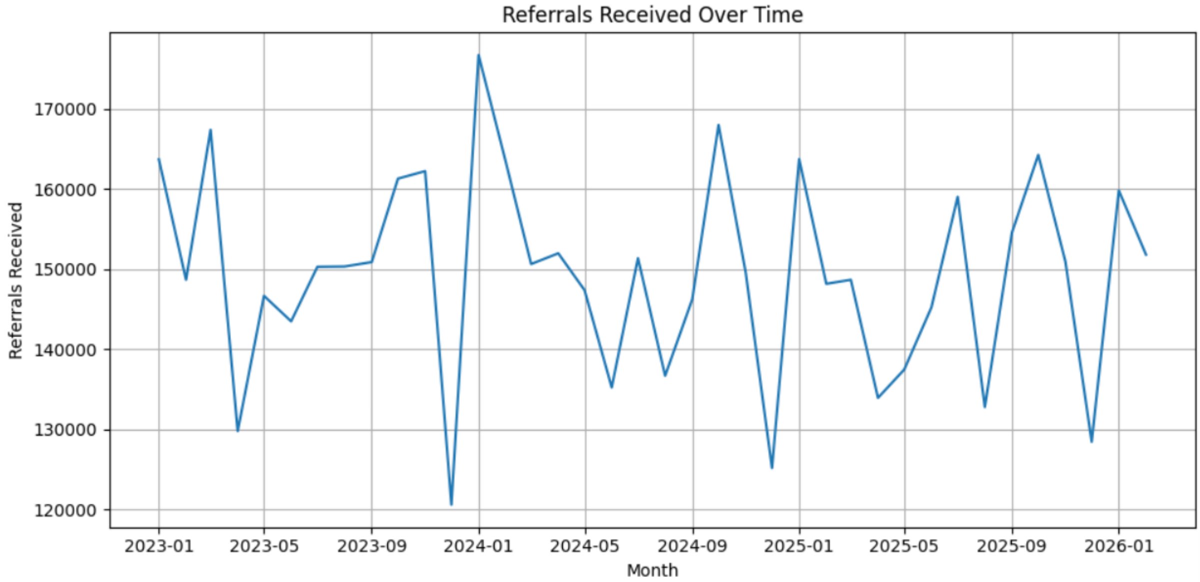
Basic Information of data:

```
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Month               38 non-null     object
 1   Referrals_Received  38 non-null     int64
```

The data type of month is changed to datatime:

```
Datatype of Month Column =  datetime64[ns]
```

Over time trend in the data:



2. Splitting the data:

Here **32 months** are used as training data and **6 months** used as testing data.

```
Training data contains 32 months up to 2025-08-01
Testing data contains 6 months from 2025-09-01
```

3. Training Model:

So, two models are trained and compared and final model is chosen.
Two models are Prophet and ARIMA.

- **Prophet:**

Prophet model is trained on the time series.

```
Prophet model trained successfully on the training data.
```

Then it is evaluated and we get:
MAE = 5678.88
RMSE = 6869.27

```
Prophet MAE on test set: 5678.88
Prophet RMSE on test set: 6869.27
```

- **ARIMA:**

Then ARIMA is trained and Evaluated:

```
Fitting ARIMA model with order (2, 1, 2) on training data.
ARIMA model fitted successfully on training data.
```

After Evaluation we get:

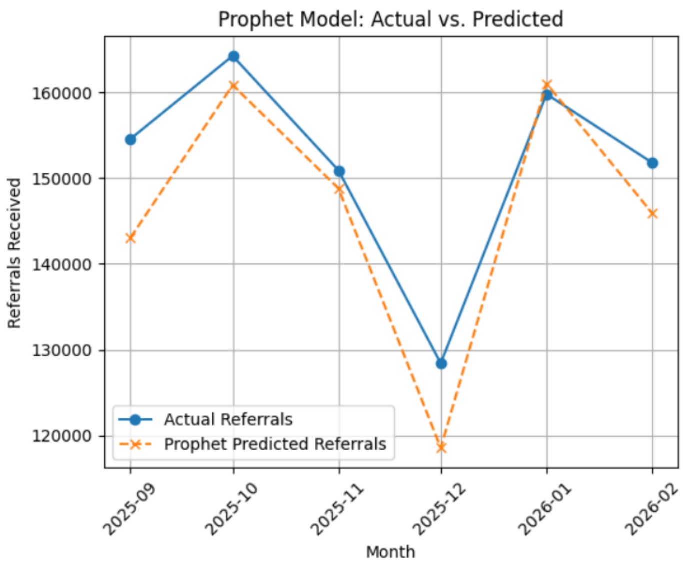
MAE = 13518.13

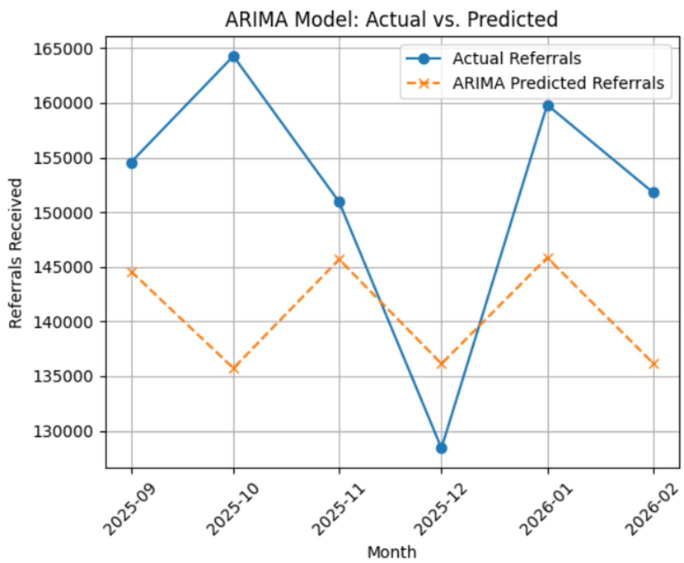
RMSE = 15499.45

```
ARIMA MAE on test set: 13518.13
ARIMA RMSE on test set: 15499.45
```

- **Comparison:**

So, now we have both models trained. Now we have to compare. We can compare with the evaluation metrices directly and also by plotting graphs.





Clearly, we see **Prophet** model **performs better** than **ARIMA**.

So, Prophet is our final model.

4. Predicting Next 3 Months:

Now, we have next 3 months prediction from our prophet forecasting model. We have Upper and Lower Confidence Intervals also.

	ds	yhat	yhat_lower	yhat_upper
38	2026-03-01	152423.840385	147953.914084	156681.602932
39	2026-04-01	126766.798169	122319.635503	131351.312711
40	2026-05-01	136546.613795	132374.933619	140985.633236

Final, graph for overall model, Predicted vs Actual and also future 3 months data points with confidence Intervals.